Closed-Loop Streaming Synthesis Bounded-Drift, Arbitrary-Length Audio–Video Generation by Treating Chunk Hand-off as Feedback Control

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Abstract

Generating arbitrarily long videos with diffusion transformers requires splitting the temporal dimension into overlapping chunks. Naive open-loop chunk-streaming accumulates exposure-bias drift: each chunk is conditioned on the previously generated latent, compounding distributional errors and causing scene collapse or grain amplification after a few hundred frames. We present **Closed-Loop Streaming Synthesis (CLSS)**, a suite of four post-denoising corrections that close the feedback loop between successive chunks without modifying the transformer weights: calibrated context re-noising, per-channel EMA-tracked AdaIN, frequency-band soft shrinkage, and a dynamic anchor bank. CLSS is implemented as a ComfyUI custom-node pipeline targeting LTX-2.3 22B (a 44 GB audio-video diffusion transformer) on consumer 16 GB VRAM hardware via GGUF 4-bit quantisation and CPU block-streaming. A two-stage pipeline generates a draft video via CLSS streaming and then refines it using a 3-step distilled-LoRA schedule at 2× spatial resolution.

Two 30-second, 3-scene audio-video experiments are reported: text-to-video (T2V) and image-to-video (I2V) generation, both using new_lf = 31, overlap = 8. T2V Stage 1 achieves boundary cosine similarities of 0.9949–0.9950 and intra-chunk similarities of 0.73–0.89; Stage 2 refinement achieves 0.984–0.993 boundary and 0.917–0.987 intra across five refinement chunks. I2V achieves comparable boundary continuity (0.946–0.950) with lower intra-chunk similarity (0.26–0.59) reflecting the more dynamic visual transitions inherent to image-conditioned generation. A cross-modal audio observation is apparent: Stage 1 chunk 1 audio coherence is -0.25 to -0.34 in T2V (no visual anchor, no audio SLB) but $+0.998$ to $+0.999$ in I2V (same audio SLB absence, but with a guide image), indicating that visual context constrains audio generation via cross-modal attention in the joint audio-video transformer. Total generation time is approximately 55–56 minutes per 30-second clip on a single 16 GB GPU.

1 Introduction

Modern video diffusion transformers operate on patchified latent representations where temporal resolution is down-scaled $8\times$ relative to pixel frames. Fitting a 30-second clip (81,840 tokens at 22×40 spatial) into a single forward pass requires very large GPUs. The standard remedy is *temporal chunking* via a Streaming Latent Buffer (SLB): generate the video in short overlapping segments, conditioning each on the previous denoised output. Section 2 reviews this mechanism and the exposure-bias problem it introduces.

CLSS closes the feedback loop between chunks by applying four lightweight post-denoising corrections that together bound the closed-loop gain below one, preventing drift from accumulating without modifying the transformer weights. The corrections are applied purely at the boundary between chunks and add negligible computational overhead.

Contributions.

1. Four closed-loop correction modules for temporal chunk streaming: calibrated context re-noising, EMA-tracked per-channel AdaIN, frequency-band soft shrinkage, and a dynamic anchor bank (Sections 3.1–3.4).
2. A linear stability analysis of the feedback loop in latent space, including an empirical per-chunk gain measurement protocol (Section 3.5).
3. A memory-efficient two-stage pipeline for LTX-2.3 22B on 16 GB VRAM using GGUF 4-bit quantisation and CPU block-streaming (Section 5).
4. ComfyUI custom nodes implementing the full pipeline with per-chunk diagnostic telemetry (Section 5.4).
5. Empirical validation on a 30-second, 3-scene audio-video generation (Section 6).
6. A reference-parity multi-modal guider for ComfyUI (split per-modality CFG, modality-isolation guidance, and STG via value-passthrough self-attention skip), including handling of ComfyUI's packed-latent sampling representation (Section 5.5).
7. A documented audit trail of two silent-failure defects (unseeded audio noise; packed-latent guidance bypass) with the telemetry that exposed them — unconditional settings dumps, per-frame origin/layout drift events, and a phase-lock repetition detector (Sections 6.2, 7.6).

2 Background

2.1 LTX-2.3 Architecture

LTX-2.3 is a 22-billion-parameter diffusion transformer for joint audio-video synthesis. It operates on a latent space with an $8\times$ temporal downscale (video VAE, causal) and a $160 \times 4\times$ audio downscale (audio VAE, 16 kHz, hop length 160, compression factor 4), yielding a video latent rate of $25/8 = 3.125$ lf/s and an audio latent rate of $16000/(160 \times 4) = 25.0$ lf/s. The model accepts interleaved video and audio tokens together with Gemma-3 text embeddings and is trained with flow-matching on the rectified-flow schedule.

2.2 Streaming Latent Buffer (SLB)

The SLB provides temporal continuity in chunk streaming. When generating chunk N, the last overlap_lf latent frames from chunk $N-1$ are prepended to the new chunk's latent, and the denoiser is conditioned on them at noise level τ_c:

$$\tilde{L}_{\text{overlap}} = (1-\tau_c)\,\hat{L}_{\text{overlap}} + \tau_c\,\varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I) \tag{1}$$

Setting $\tau_c = 0$ yields maximal continuity but maximal drift risk; $\tau_c = 1$ decouples the chunks entirely. In LTX's framework this is implemented as `VideoConditionByLatentIndex` with strength $= 1 - \tau_c$.

2.3 Exposure Bias in Streaming Diffusion

Let $\hat{L}_N$ be the denoised latent of chunk N and $f_\theta(\cdot \mid \hat{L}_{N-1})$ the denoising operator conditioned on the previous chunk. In open-loop streaming:

$$\hat{L}_N = f_\theta(\hat{L}_{N-1} + \varepsilon_N) \tag{2}$$

The transformer was trained on latents drawn from the data distribution, but at inference it receives denoised latents from prior chunks, which lie off-manifold relative to its training distribution. Any distributional shift $\delta_N = \hat{L}_N - L_N^*$ (where L_N^* is the "on-manifold" latent for the same content) is in general amplified by the open-loop gain of the denoising process. Formally, if $g = \|\partial f_\theta / \partial L\|$ evaluated at L_{N-1}^*, then $\|\delta_{N+1}\| \approx g\|\delta_N\|$. When $g > 1$ this gives $\|\delta_N\| \sim g^N \to \infty$.

In practice g is not a global constant: it depends on the noise level schedule, step count, guidance strength, and the spectral content of the latent. Section 3.5 describes how g is estimated empirically per chunk via a perturbation measurement.

3 Closed-Loop Streaming Synthesis

CLSS adds four closed-loop correction modules that together bound the error accumulation by ensuring $\rho_{\text{closed}} = g \cdot (1-\beta) \cdot (1-\tau_c) < 1$.

3.1 Calibrated Context Re-Noising

The overlap latent fed as context to chunk N is re-noised to level τ_c via Eq. (1). This forces the denoiser to actively re-project the boundary context onto the data manifold rather than accepting it verbatim, partially correcting accumulated drift in the SLB.

In practice $\tau_c = 0.05$ provides a strong temporal constraint (boundary cosine similarity > 0.97) while allowing enough distributional repair to prevent drift accumulation. Larger values ($\tau_c = 0.20$, as suggested by some prior work) loosen the constraint too much for 22B GGUF streaming, causing visible motion discontinuities.

3.2 EMA-Tracked Per-Channel AdaIN

A slow exponential moving average (EMA) with rate $\lambda \approx 0.10$ tracks per-channel mean μ_c and standard deviation σ_c across chunks:

$$\mu_c^{(N)} = (1-\lambda)\,\mu_c^{(N-1)} + \lambda\,\hat{\mu}_c^{(N)} \tag{3}$$
$$\sigma_c^{(N)} = (1-\lambda)\,\sigma_c^{(N-1)} + \lambda\,\hat{\sigma}_c^{(N)} \tag{4}$$

where $\hat{\mu}_c^{(N)}, \hat{\sigma}_c^{(N)}$ are the per-channel statistics of chunk N's new frames. After generation, each chunk's new frames are blended toward the EMA reference via AdaIN with blend factor β:

$$\tilde{L}_N = (1-\beta)\,L_N + \beta\ \mathrm{AdaIN}(L_N \to \mu^{(N-1)}, \sigma^{(N-1)}) \tag{5}$$

This suppresses fast per-chunk statistical drift while allowing slow intentional scene evolution (lighting changes, colour grading) to pass through.

Two safety caps prevent over-correction:

- **Sigma max drift**: the EMA σ_c is clamped to at most $(1+\Delta) \times \sigma_c^{(0)}$ (default $\Delta = 0.05$), preventing AdaIN from amplifying late-chunk residual noise.
- **AdaIN max amplification**: per-channel upward scaling is capped at $1.2\times$ the current chunk's std (default), preventing AdaIN from boosting noise in channels that have become quiet.

3.3 Frequency-Band Soft Shrinkage

The latent is decomposed into $B = 4$ frequency bands via 3-D FFT along the (frame, height, width) axes. Each band b receives a continuous gain:

$$g_b^{(c)} = \min\left(1,\ \left(\frac{E_{\mathrm{ref},b}^{(c)}}{E_b^{(c)}}\right)^{\gamma_b}\right) \in (0,1] \tag{6}$$

where $E_b^{(c)}$ is the mean squared amplitude of band b in channel c, $E_{\mathrm{ref},b}^{(c)}$ is the corresponding EMA reference, and γ_b is the per-band shrinkage exponent.

Only bands with excess energy relative to their EMA are attenuated. Bands with $\gamma_b = 0$ (the DC band) are never attenuated. Default exponents: $\boldsymbol{\gamma} = (0.0, 0.05, 0.2, 0.3)$, which are conservative enough to preserve scene structure while suppressing temporal flicker.

The EMA is updated with the *corrected* (post-gain) energies, not the raw ones, so the reference tracks intentional scene content rather than accumulated error.

3.4 Dynamic Anchor Bank

A small bank of $M \leq 8$ keyframe anchors provides long-range visual identity conditioning. The bank is seeded from the first frame of chunk 1 and updated when two consecutive chunks trigger a scene-change signal (cosine similarity below threshold $\rho_a = 0.78$ for $P = 2$ consecutive chunks). Additionally, a *forced insertion* every $F_a = 5$ chunks prevents bank stagnation on visually stable sequences where the similarity threshold is never triggered.

Anchors are retrieved via cosine similarity and injected as `VideoConditionByKeyframeIndex` at frame index 0 of the current chunk. Anchors too similar to the current overlap (sim > 0.85) are skipped to avoid duplicating the SLB conditioning. LRU eviction discards the least recently used anchor when the bank is full.

3.5 Stability Analysis

The AdaIN blend (Section 3.2) and context re-noising (Section 3.1) together reduce the magnitude of any per-chunk latent perturbation by a factor $(1-\beta)(1-\tau_c)$ before it reaches the next chunk's denoiser. This gives the *loop gain*:

$$\rho_{\text{loop}} = (1-\beta)\cdot(1-\tau_c) \tag{7}$$

The combined closed-loop gain, accounting for the model's open-loop amplification g, is:

$$\rho_{\text{closed}} = g \cdot \rho_{\text{loop}} \tag{8}$$

Bounded drift requires $\rho_{\text{closed}} < 1$. With $\beta = 0.40$ and $\tau_c = 0.05$, $\rho_{\text{loop}} = 0.57$, providing $\approx 43\%$ gain reduction per chunk before accounting for g.

Note that Eq. (7) is a first-order approximation that treats the two corrections as independent multiplicative attenuators. In practice g is not constant: it depends on the step count, guidance strength, noise schedule, and the spectral properties of the current latent. The analysis therefore provides a stability condition rather than a precise quantitative bound. The frequency-band shrinkage (Section 3.3) provides additional spectral attenuation beyond ρ_{loop}, especially in high-frequency bands, but is not included in the linear gain model because its contribution is content- and chunk-dependent.

Hyperparameter Choices. $\beta = 0.40$ is the largest value that preserves scene-change responsiveness in practice: at $\beta \geq 0.5$ slow colour shifts accumulate due to excessive EMA anchoring. $\tau_c = 0.05$ is a conservative re-noising level that maintains high boundary similarity (> 0.97) while partially correcting SLB distribution shift; larger values ($\tau_c \geq 0.20$) cause visible motion discontinuities on 22B streaming. Together they yield $\rho_{\text{loop}} = 0.57$, which empirically provides adequate margin given that g appears to be below ≈ 1.07 for the tested configuration (Section 7). $\lambda = 0.10$ (EMA rate) was chosen to respond to gradual scene evolution without over-weighting individual chunks; $\boldsymbol{\gamma} = (0.0, 0.05, 0.2, 0.3)$ are conservative exponents that attenuate only bands with clear excess energy.

Per-Chunk Gain Measurement. g is measured empirically by running a second denoising pass with a slightly perturbed overlap latent $\tilde{L}_{\text{overlap}} + \delta$ ($\|\delta\|_F =$

$\epsilon \, \|\tilde{L}_{\text{overlap}}\|_F$, $\epsilon = 0.01$) and comparing the output perturbation to the input:

$$g_{\text{SLB}} = \frac{\|\Delta^{\text{SLB}}_{\text{output}}\|_F}{\|\delta\|_F} \tag{9}$$

where $\Delta^{\text{SLB}}_{\text{output}}$ is restricted to the last overlap_lf new frames—the portion that becomes the next chunk's SLB. This measures sensitivity of the boundary output to boundary input perturbations, which is the relevant quantity for SLB-mediated drift propagation. It does not capture drift arising from other channels (e.g., prompt conditioning or attention to non-overlap frames), so g_{SLB} is a lower bound on the true open-loop gain.

4 Algorithm

Algorithm 1 summarises the per-chunk CLSS procedure.

Algorithm 1 CLSS per-chunk generation

Require: Config: $\tau_c, \beta, \lambda, \boldsymbol{\gamma}$; state: SLB, EMA, anchor bank

```
L_ov ← SLB.read()                                        ▷ may be empty for chunk 0
L̃_ov ← (1 − τ_c) L_ov + τ_c ε                            ▷ Eq. (1); Sect. 3.1
A ← anchor_bank.retrieve(top-m by cosine sim)            ▷ Sect. 3.4
L_N ← Denoise(L̃_ov, A, prompt_N)
L_N^new ← L_N[overlap_lf :]                              ▷ drop overlap region
L_N^new ← AdaIN_blend(L_N^new, EMA, β)                   ▷ Sect. 3.2
L_N^new ← band_soft_shrink(L_N^new, band_EMA, γ)         ▷ Sect. 3.3
EMA.update(L_N^new, λ); band_EMA.update(L_N^new, λ)
anchor_bank.update(last frame of L_N^new)                ▷ Sect. 3.4
SLB.push(L_N^new[−overlap_lf :])
return L_N^new
```

5 Implementation

5.1 Memory Budget for 16 GB VRAM

LTX-2.3 22B in BF16 occupies ≈ 44 GB. Three complementary strategies fit it within 16 GB VRAM:

1. **GGUF 4-bit quantisation** (Q4_K_S): reduces the on-disk size to ≈ 11 GB. Dequantization to BF16 at load time requires ≈ 44 GB of pinned CPU RAM.
2. **CPU block-streaming**: all transformer blocks reside in pinned CPU RAM; only 2 blocks are resident on GPU at a time (≈ 5 GB model weights on GPU), with the next block pre-fetched asynchronously via a dedicated CUDA stream.
3. **Temporal chunking (CLSS)**: latent memory is $O(\text{overlap\_lf})$ instead of $O(\text{total\_frames})$. Processed chunk latents are immediately offloaded to CPU after each chunk.

Combined peak VRAM budget at 512×768 resolution:

Component	Peak VRAM
Model blocks ($2 \times$ BF16, streaming)	≈ 5 GB
Activations + latents (per chunk)	≈ 4 GB
VAE / text encoder (loaded/freed)	≈ 2 GB
Total peak	≈ 11 GB

5.2 Two-Stage Pipeline

Stage 1 (Open-resolution streaming). CLSS streaming generates the full video at a coarse latent resolution ($H_L \times W_L$ latent grid, typically 11×20 for 704×512 output). Each chunk runs the full 20–30-step LTX scheduler. CLSS corrections are applied between chunks as per Algorithm 1.

Stage 2 (Distilled-LoRA refinement). Stage 1 operates at 704×512 output resolution and 20 denoising steps, which is sufficient for temporal structure and scene content but produces visible quantisation artefacts and reduced texture detail, particularly after GGUF 4-bit dequantisation at the low-frequency latent scale. Stage 2 addresses this by upsampling and refining the output.

The Stage 1 output is spatially upsampled $2\times$ by `LTXVLatentUpsampler`, yielding a 22×40 latent grid corresponding to 1408×1024 pixel resolution. Stage 2 then refines the full video in temporal chunks using a 3-step distilled-LoRA schedule with initial noise level $\sigma_0 \approx 0.909$. Each chunk's new frames are seeded from the upscaled Stage 1 latent with σ_0-scaled noise:

$$x_{\text{start}} = \sigma_0 \, \varepsilon + (1 - \sigma_0) \, L_N^{\text{S1}} \tag{10}$$

This formulation anchors Stage 2 to the coarse temporal structure produced by Stage 1—preserving scene layout, motion trajectories, and chunk boundaries established by CLSS—while allowing the 3-step refinement to add local high-frequency detail. Stage 1 alone could in principle be generated at $2\times$ resolution, but this would quadruple token count and exceed VRAM budget under CPU block-streaming.

Key differences from Stage 1:

- AdaIN and spectral-shrinkage corrections are *disabled*: Stage 2 refinement is structurally anchored to the S1 latent, so distributional drift correction is unnecessary.
- A global pre-generated noise field is sliced per chunk (`_SlicedNoise`) to prevent grain seams at chunk boundaries—a single coherent noise tensor ensures that the per-chunk noise samples tile seamlessly when concatenated.
- Audio is refined in parallel with the SLB at τ_c using the Stage 2 audio tail as context.

5.3 Audio Streaming

Audio and video share the same temporal chunk boundaries. Audio length accounting is anchored to *pixel time*: the causal video VAE maps a chunk of new_lf latent frames to $(\text{new\_lf} - 1) \cdot 8 + 1$ pixel frames for the first chunk of a sequence and $\text{new\_lf} \cdot 8$ pixel frames for every continuation chunk. The audio rate is taken from the chunk template, $r = \text{new\_af}/\big((\text{new\_lf} - 1) \cdot 8 + 1\big)$ audio frames per pixel frame, and the SLB length follows the same rate:

$$\text{overlap\_af} = \text{round}(\text{overlap\_lf} \cdot 8 \cdot r) \tag{11}$$

With overlap_lf = 8 and the 13-lf/102-af template this gives overlap_af = 67. (An earlier per-chunk accounting ignored the causal first-frame discount, undercounting every continuation chunk by ≈ 0.28 s and desynchronising audio from video by ≈ 2 s over seven chunks; the pixel-time ledger above eliminates the drift exactly.) The effective video overlap is additionally clamped to new_lf: the SLB can never carry more frames than a chunk produces, and unclamped values were measured to silently discard motion at every seam.

Drift-capped energy anchoring. Un-anchored streaming audio is a divergent loop in both directions: runs were measured with RMS growing +67% and low-mid spectral bands tripling over ten chunks. A per-chunk anchor corrects the new audio's global RMS and per-channel DC toward slowly-moving EMA targets whose drift from the chunk-1 origin is hard-capped at $\pm\Delta_\sigma$ (the same ±5% used for the video EMA), before the chunk's tail is fed forward as the next SLB/reference. Anchoring *before* feed-forward is what makes the audio SLB safe at any sequence length. A fixed-forever target was tried first and caused repetition (the model re-rendered a stabilised loop); the capped EMA restores bounded evolution. On multi-scene runs the anchor is re-baselined at scene changes; this is measurably a ratchet (each re-baseline legitimises accumulated loudness creep: $0.599 \rightarrow 0.699 \rightarrow 0.823$ over three scenes in one run) and remains a known limitation.

Reference-audio conditioning. Each continuation chunk also receives a reference-audio window (`ref_audio` tokens at negative temporal RoPE) taken from a rolling tail of the accumulated output. Its length is decoupled from the video overlap (default 67 af ≈ 2 s; an implicit coupling to overlap_af starved it at small overlaps and measurably collapsed within-chunk audio coherence from 0.91 to 0.27). A small noise fraction is blended into the reference, ramping 0.05/chunk and capped at 0.10 (an earlier hard-coded 0.35 cap injected audible hiss).

Context envelope flattening. The fed-forward audio context (SLB tail and reference window) is optionally flattened in loudness: each frame is rescaled toward the window-mean RMS with per-frame gain clamped to $[0.6, 1.25]$. Motivation is a measured metronome: the audio energy peak locked to the same frame (102/109) for eight consecutive chunks — the crescendo in each fed-forward tail re-seeded the same loudness arc every chunk. Flattening the *context only* (output audio is untouched) broke the phase lock in A/B runs. The gain cap is asymmetric by design: an unbounded first version amplified the noise floor of quiet frames ($\times 2.98$ observed) and injected persistent hiss.

Boundary smoothing. At each audio SLB boundary, a linear blend over ± 2 audio latent frames is applied in latent space before VAE decoding. This eliminates click artifacts that arise from the abrupt splice between the overlap tail and the new chunk, which would otherwise be visible as discontinuities in the spectrogram.

5.4 ComfyUI Node Integration

Five custom nodes implement the full pipeline inside ComfyUI:

1. **CLSSConfig**: exposes the CLSS hyperparameters worth exposing (τ_c, β, overlap); validated internals are fixed.
2. **CLSSScenePrompts**: multi-scene text input via `---` delimiters; Gemma-enhanced per scene. Output is a flat CONDITIONING list with one entry per scene, proportionally assigned to chunks. With one prompt per chunk this doubles as a per-chunk *object-identity* anchor (see Section 8).
3. **CLSSStreamingSampler**: Stage 1 streaming sampler. Accepts any ComfyUI GUIDER/SAMPLER/SIGMAS/NOISE/LATENT combination. Emits an unconditional settings dump, per-chunk telemetry, end-of-run trend summaries with per-chunk localisation, per-frame origin/layout drift event tables, and a phase-lock repetition detector.
4. **CLSSStage2**: Stage 2 refinement sampler with consistent per-chunk noise slicing and a decoupled audio-refinement pass.
5. **CLSSAVGuiderV2**: reference-parity multi-modal guider (Section 5.5). Its predecessor (**CLSSAVGuider**, a `sampler_cfg_function` patch) is retained but inert on current ComfyUI and warns accordingly.

All stochastic inputs are seeded: Stage-1 video *and* audio noise are pre-generated as full-sequence fields sliced per chunk (distinct yet coherent noise per chunk, bit-reproducible runs), and every fallback noise path uses a generator derived deterministically from the seed and chunk position.

5.5 Multi-Modal Guidance and the Packed-Latent Pitfall

The reference pipeline guides each modality independently (`MultiModalGuider`):

$$\text{pred} = c + (s_{\text{cfg}} - 1)(c - u) + s_{\text{stg}}\,(c - p) + (s_{\text{mod}} - 1)(c - m) \tag{12}$$

per modality, followed by a rescale pred $\mathrel{*}= r\,\sigma_c/\sigma_{\text{pred}} + (1 - r)$, where u is the text-unconditional prediction, m the modality-isolated prediction (both audio$\leftrightarrow$video cross-attentions skipped), and p the STG-perturbed prediction. The STG perturbation skips *self*-attention in designated blocks (block 28 for LTX-2.3) by replacing the attention output with the raw value projection (out $= W_V x$, gating and output projection unchanged) — a value-passthrough, not a zero-out. Defaults follow the reference: video cfg 4–4.5, audio cfg 7, modality 3, STG 1 for both modalities, rescale 0.7; four transformer passes per step.

The pitfall. ComfyUI's `CFGGuider.sample()` packs nested audio-video latents into a *single flat* $[B, 1, N_v + N_a]$ tensor before sampling and unpacks only after. Any guider that detects the AV case with `isinstance(x, NestedTensor)` inside `predict_noise` — as both of our guiders originally did — silently falls back to plain shared CFG on every step. The failure produced no error and no log line; it was exposed by two independent signals: the guider's one-time diagnostic prints appeared in no console log, and per-step wall-clock time was invariant (8.0 s/it) across guidance configurations whose pass counts should have differed by 2×. The fix captures the per-modality shapes in `sample()` before packing, detects the packed representation in `predict_noise`, unpacks each pass output, applies Eq. (12) per modality, and repacks. Verified in situ: `passes/step=4`, non-zero STG/modality difference norms, and step time at its honest cost (8.0 → 17.3 s/it on 13-lf chunks). Two engineering rules follow: guidance paths must announce which branch is active, loudly and unconditionally; and a guidance change that does not change step time did not happen.

6 Experiments

Two generation experiments are reported, both using the same CLSS configuration but differing in the first-frame conditioning: Experiment 1 uses text-to-video (T2V) generation with no image prior; Experiment 2 uses image-to-video (I2V) generation conditioned on a single guide frame ($1 \times 11 \times 20$ latent).

6.1 Shared Configuration

Table 1: Shared configuration for both experiments.

Parameter	Value
Model	LTX-2.3 22B (Q4_K_S GGUF)
Resolution	704×512 (Stage 1), 1408×1024 (Stage 2)
Frames	93 latent $= \approx 30$ s @ 25 fps
Stage 1 chunks	3, new_lf = 31, overlap = 8
Stage 1 steps	20
Stage 2 chunks	5, fpc = 21, overlap = 8
Stage 2 steps	3 (distilled)
τ_c	0.05
β	0.40
λ	0.10
Δ_σ	0.05
F_a	5
$\boldsymbol{\gamma}$	(0.0, 0.05, 0.2, 0.3)
Video CFG	4.5, STG 1.0, modality 3.0 *(intended; see §6.2)*
Audio CFG	7.0, STG 1.0 *(intended; see §6.2)*
Scenes	3 (one per Stage 1 chunk)
VRAM (GPU)	15.61 GB (RTX-class)
Exp. 1 first-frame	Text only (T2V)
Exp. 2 first-frame	Guide image, latent $[1, 128, 1, 11, 20]$ (I2V)

6.2 Erratum: Guidance Configuration (July 2026 Audit)

A post-publication audit established three facts that qualify the experiments below. **(i) The guidance stack was not active.** Owing to the packed-latent pitfall (Section 5.5), every ComfyUI run in this section executed plain shared CFG at the video scale — no split audio CFG, no modality guidance, no STG, no per-modality rescale — regardless of the values in Table 1. The continuity results below therefore measure the CLSS correction stack under *plain CFG*, which if anything understates the corrected pipeline. **(ii) Audio noise was unseeded.** Stage-1 audio noise was drawn from the global RNG rather than the seed; runs were not bit-reproducible, and audio (via joint attention, also video) differed across same-seed runs. Both defects are fixed; Stage-1 noise for both modalities is now pre-generated from the seed and sliced per chunk. **(iii) The anchor bank was diagnostic-only.** The retrieval-injection path of Section 3.4 exists in the reference library but was not wired into the ComfyUI sampler; in these experiments the bank contributed telemetry (identity tracking, forced insertions) but no conditioning.

The first run with the full guidance stack verified active (2026-07-11; 3 chunks, i2v, 13 lf) was the most stationary measured to date: audio RMS plateaued at 0.49–0.51

(versus 0.74–0.99 in unguided runs), per-chunk spectral ratios stayed within 0.72–1.19 of the chunk-1 reference (the flattest spectral drift observed), within-chunk audio coherence reached 0.95–0.99, and video boundary similarities held at 0.954–0.959 — at the honest cost of four transformer passes per step.

6.3 Experiment 1: Text-to-Video (T2V)

6.3.1 Stage 1 Results

Table 2 reports per-chunk metrics. Chunk 1 is the reference; subsequent chunks compare against the anchor bank entry seeded from chunk 1's last frame (frame 30).

Table 2: T2V Stage 1 per-chunk metrics. Identity_sim for chunk 2 nearly equals boundary_sim because both compare the same pair of frames (anchor at frame 30 vs first frame of chunk 2), confirming SLB and anchor bank agree.

Chunk	Boundary sim↑	Intra sim↑	Identity sim↑	ρ_{loop}
1/3	N/A	0.7333	1.0000 (ref)	0.570
2/3	0.9949	0.7375	0.9949	0.570
3/3	0.9950	0.8861	0.7224	0.570

Boundary similarities (0.9949, 0.9950) are nearly identical across both transitions, confirming that the SLB with $\tau_c = 0.05$ reliably locks the visual hand-off regardless of chunk content. Intra-chunk similarities (0.73–0.89) reflect the natural within-chunk variation expected for new_lf = 31 spanning approximately 10 s per chunk with three distinct scenes. The identity_sim drop to 0.7224 by chunk 3 reflects two accumulated scene transitions over 62 latent frames, consistent with intended multi-scene content rather than model drift.

Table 3 shows AdaIN corrections and spectral band gains. Chunk 1 receives no correction (EMA uninitialised; it is the reference). Chunk 2 receives a small upward mean shift (+0.007) and moderate std lift (+0.044). By chunk 3, the EMA has largely converged, and spectral shrinkage takes over: band 3 gain falls to 0.9377, indicating that scene 3's high-frequency latent energy had grown $\approx 6.7\%$ above the EMA reference established by scenes 1–2.

Table 3: T2V Stage 1 AdaIN corrections and band gains. Band 0 is the DC band ($\gamma_0 = 0$, gain$\equiv 1$).

Chunk	$\Delta\mu$	$\Delta\sigma$	g_0	g_1	g_2	g_3
1	+0.000	+0.000	1.0000	1.0000	1.0000	1.0000
2	+0.007	+0.044	1.0000	0.9997	0.9917	0.9946
3	+0.003	+0.007	1.0000	0.9962	0.9659	0.9377

The anchor bank retains one entry throughout (bank_size= 1, frame 30). Chunk 2 reaches last_frame_max_sim= 0.743 (streak= 1), but persistence requires $P = 2$ consecutive sub-threshold chunks before a new anchor is committed. Chunk 3 returns above threshold (sim= 0.802, streak resets), so no second anchor is committed.

6.3.2 Stage 2 Results

Table 4: T2V Stage 2 per-chunk metrics (5 chunks, fpc = 21). Identity_sim compares against Stage 2 chunk 1's first frame; the 0.41–0.52 plateau reflects 3 distinct scenes rather than model instability.

Chunk	Frames (lat)	Boundary sim↑	Intra sim↑	Identity sim↑
1/5	0–21	N/A	0.4797	1.0000 (ref)
2/5	21–42	0.9928	0.9870	0.4798
3/5	42–63	0.9839	0.9749	0.5204
4/5	63–84	0.9856	0.9355	0.4120
5/5	84–93	0.9904	0.9174	0.4310

Stage 2 boundary similarities (0.984–0.993) confirm smooth transitions at all four chunk boundaries. Intra-chunk similarities are substantially higher than Stage 1 (0.917–0.987 for chunks 2–5 vs 0.73–0.89), as Stage 2 is anchored to the S1 upscaled latent and requires only 3 refinement steps. The low first-chunk intra similarity (0.4797) reflects the absence of SLB overlap for Stage 2 chunk 1 and the inherent scene variation already present in Stage 1 scene 1.

The full refined video latent has shape $[1, 128, 93, 22, 40]$, mean= 0.0293, std= 0.9970, no NaN/Inf. The assembled audio std (0.916) is substantially higher than the Stage 1 audio std (0.650), confirming that the distilled refinement actively reshapes the audio distribution.

6.3.3 Audio Coherence

Table 5: T2V within-chunk audio similarity (cosine, three sequential segment pairs).

Stage	Chunk	Seg 1→2	Seg 2→3
S1	1/3	−0.254	−0.337
S1	2/3	+0.940	+0.814
S1	3/3	+0.990	+0.987
S2	1/5	−0.469	+0.901
S2	2/5	−0.044	+0.374
S2	3/5	+0.659	+0.305
S2	4/5	+0.956	+0.940
S2	5/5	+0.911	+0.913

Stage 1 chunk 1 audio is strongly anti-correlated ($-0.254 \to -0.337$), a consequence of generating from pure noise with no audio SLB and no strong visual prior. Once the audio SLB activates at chunk 2, coherence jumps to $0.940 \to 0.814$ and reaches $0.990 \to 0.987$ by chunk 3.

Stage 2 audio follows the same causal pattern: the first two chunks show mixed coherence (reflecting that S2 inherits the incoherent S1 chunk 1 audio), while chunks 4–5 reach 0.91–0.96 as the audio SLB accumulates refined context.

6.4 Experiment 2: Image-to-Video (I2V)

In this experiment a single guide image is encoded to a latent of shape $[1, 128, 1, 11, 20]$ and provided as first-frame conditioning to Stage 1 chunk 1. All CLSS hyperparameters are identical to Experiment 1.

6.4.1 Stage 1 Results

Table 6: I2V Stage 1 per-chunk metrics.

Chunk	Boundary sim↑	Intra sim↑	Identity sim↑	ρ_{loop}
1/3	N/A	0.2649	1.0000 (ref)	0.570
2/3	0.9495	0.5538	0.9495	0.570
3/3	0.9464	0.5904	0.7638	0.570

Boundary continuity remains high (0.9495, 0.9464), demonstrating CLSS correctness under image conditioning. Intra-chunk similarities (0.26–0.59) are lower than T2V (0.73–0.89): this is expected, as I2V mode generates video that must travel from a specific fixed first frame through the scene narrative, producing more within-chunk visual change than unconstrained T2V. Chunk 1's intra similarity of 0.2649 (vs 0.7333 in T2V) directly reflects this: the model starts at the guide image and moves significantly away from it over 31 latent frames.

Table 7: I2V Stage 1 AdaIN corrections and band gains.

Chunk	$\Delta\mu$	$\Delta\sigma$	g_0	g_1	g_2	g_3
1	+0.000	+0.000	1.0000	1.0000	1.0000	1.0000
2	−0.009	−0.063	1.0000	0.9931	0.9921	0.9918
3	+0.002	−0.168	1.0000	0.9945	0.9922	0.9945

The AdaIN correction in chunk 2 has a downward std direction ($\Delta\sigma = -0.063$) compared to the upward direction in T2V (+0.044): the image-conditioned chunk 1 produces a higher std than the subsequent text-only chunks, so AdaIN here compresses rather than expands. Chunk 3 requires the largest std correction of either experiment (−0.168), pulling the raw pre-correction std of 1.12 down to 0.95; the band 3 shrinkage gain (0.9945) is mild here because spectral energy has not grown as strongly as in T2V.

The anchor bank grows to size 2 by chunk 3: chunk 2 triggers a scene-change streak (sim= 0.7343, streak= 1), and chunk 3 commits a second anchor at frame 92 (bank_frame_ids= [30, 92]), growing the bank from 1 to 2.

6.4.2 Stage 2 Results

Table 8: I2V Stage 2 per-chunk metrics. Identity_sim compares against Stage 2 chunk 1's first frame (the guide-image frame); the negative values for chunks 3–5 reflect that those scenes are substantially different from the opening image.

Chunk	Frames (lat)	Boundary sim↑	Intra sim↑	Identity sim
1/5	0–21	N/A	−0.020	1.0000 (ref)
2/5	21–42	0.9854	0.6612	+0.095
3/5	42–63	0.8342	0.7772	−0.179
4/5	63–84	0.9897	0.5971	−0.202
5/5	84–93	0.9863	0.8761	−0.200

Stage 2 boundary continuity is maintained at four of five boundaries (0.985–0.990), with one lower-similarity transition at chunk 3 (0.834). The Stage 2 chunk 3 boundary covers the mid-sequence scene transition where Stage 1 audio was also less coherent (Section 6.4.3), suggesting a localised region of lower-quality Stage 1 generation that Stage 2 partially corrects but cannot fully eliminate.

Stage 2 chunk 1 has an intra similarity of −0.020 (nearly zero), much lower than T2V's 0.480. This reflects that I2V Stage 1 chunk 1 already had very low intra similarity (0.265), and Stage 2 chunk 1 covers exactly that region (latent frames 0–21). The negative value indicates that the first and last frames of this refined chunk are essentially uncorrelated—the refined video transitions dramatically within the first 6.7 seconds, consistent with a video that begins at a specific guide image and moves through its first scene.

The identity_sim values for chunks 3–5 (−0.18 to −0.20) are substantially more negative than T2V (+0.41 to +0.52), because in I2V mode the Stage 2 reference frame is the guide image, which is by construction very different from the content of scenes 2 and 3.

6.4.3 Audio Coherence

Table 9: I2V within-chunk audio similarity (cosine, three sequential segment pairs).

Stage	Chunk	Seg 1→2	Seg 2→3
S1	1/3	+0.998	+0.999
S1	2/3	+0.984	−0.824
S1	3/3	+0.992	+0.811
S2	1/5	+0.980	+0.975
S2	2/5	+0.974	+0.890
S2	3/5	+0.163	+0.175
S2	4/5	+0.919	+0.973
S2	5/5	+0.967	−0.289

In I2V mode, Stage 1 chunk 1 audio is highly coherent ($+0.998 \rightarrow +0.999$), in stark contrast to T2V chunk 1 ($-0.254 \rightarrow -0.337$). This difference is significant: both experiments have no audio SLB at chunk 1. The sole difference is the guide image, which provides a strong visual anchor via cross-modal attention in LTX-2.3's joint audio-video transformer. This demonstrates that audio coherence in chunk 1 depends not only on the audio SLB but also on the strength of the visual context conditioning.

Stage 1 chunk 2 audio shows a within-chunk drop: the first segment pair is coherent (+0.984) but the second is anti-correlated (−0.824). This occurs at the second scene boundary, where the model transitions to new scene content partway through chunk 2; the joint audio-video denoiser appears to handle the video transition but produce momentarily incoherent audio mid-chunk.

In Stage 2, the I2V audio is highly coherent in chunks 1–2 (inheriting the coherent S1 audio) but shows a collapse at chunk 3 ($+0.163 \rightarrow +0.175$), which corresponds to the audio region seeded from S1 chunk 2's incoherent second half (−0.824). Chunks 4–5 recover to 0.92–0.97 once the Stage 2 audio SLB has accumulated sufficient refined context.

6.5 Computational Overhead

Table 10: Wall-clock timing on a 16 GB GPU (RTX-class, CPU block-streaming). T2V and I2V timings are nearly identical; token counts, not conditioning mode, dominate iteration cost.

Stage	Chunk	Frames (new+SLB)	Tokens	Time
S1 T2V	1/3	31+0	27,280	$\approx$ 6.9 min
S1 T2V	2–3	31+8	34,320	$\approx$ 8.8 min each
S1 I2V	1/3	31+0 (+1 guide)	28,160	$\approx$ 7.0 min
S1 I2V	2–3	31+8	34,320	$\approx$ 8.9 min each
S2	1/5	21+0	18,480	$\approx$ 5.0 min
S2	2–4	21+8	25,520	$\approx$ 7.4 min each
S2	5/5	9+8	14,960	$\approx$ 4.1 min

Total generation time: T2V $\approx$ 24.6 min (S1) + 30.6 min (S2) = **55 min**; I2V $\approx$ 24.9 min (S1) + 31.3 min (S2) = **56 min**. Both modes produce approximately 30 seconds of audio-video in under one hour on a single 16 GB GPU. The per-iteration cost is determined by token count rather than conditioning mode: Stage 2 chunk 1 (18,480 tokens) costs $\approx$ 100 s/it vs $\approx$ 147 s/it for 25,520-token chunks.

7 Discussion

7.1 Effect of Chunk Length

Both experiments used new_lf = 31 ($\approx$ 10 s per chunk at 25 fps), exceeding the 21-frame safe limit established in earlier open-loop testing. With CLSS corrections fully active, T2V achieves Stage 1 boundary similarities of 0.9949–0.9950 and I2V achieves 0.9464–0.9495: both well above the values obtainable without corrections. Spectral shrinkage applies progressively stronger corrections in later chunks (T2V band 3 gain= 0.938 at chunk 3; I2V band 3 gain= 0.994 at chunk 3, with the remaining correction handled by AdaIN), demonstrating that CLSS widens the practical chunk-length operating range beyond what open-loop streaming allows.

The I2V experiment's lower boundary similarities (0.946–0.950 vs T2V's 0.995) are consistent with the I2V generation producing more dynamic visual transitions: the model moves from a specific fixed frame toward the target scene, creating larger inter-chunk visual change at boundaries.

7.2 Identity Tracking and the Anchor Bank Fix

The repaired identity_sim metric reveals a qualitatively new observation: chunk 2's identity_sim (0.9949) equals its boundary_sim exactly. This is not a coincidence—it is a *self-consistency check*. The anchor was seeded from chunk 1's last frame; the SLB passes that same frame as the first latent frame of chunk 2's conditioning. Therefore the anchor query and the boundary query both compare the same source frame (chunk 1 frame 30) against the same target frame (chunk 2 frame 0). Seeing both metrics agree validates that the anchor bank and the SLB are operating on the same reference, and that the tau_c re-noising at the boundary introduces only minimal distributional shift ($\leq 0.5\%$ similarity loss).

7.3 Audio Coherence and the Role of Visual Context

The two experiments together reveal that audio coherence in chunk 1 is not determined solely by the presence of an audio SLB, but also by the strength of the visual context.

In T2V (Experiment 1), Stage 1 chunk 1 audio is anti-correlated ($-0.254 \rightarrow -0.337$) despite 20 denoising steps over 253 audio latent frames. In I2V (Experiment 2), with the same step count and the same absence of an audio SLB, chunk 1 audio reaches $+0.998 \rightarrow +0.999$. The sole conditioning difference is the guide image. In LTX-2.3's joint audio-video transformer, visual tokens and audio tokens share the same self-attention layers; a strong single-frame visual anchor therefore constrains the audio generation implicitly through cross-modal attention, preventing the audio from wandering incoherently across the chunk.

This suggests that the T2V audio incoherence at chunk 1 arises from the absence of *any* strong conditioning anchor—neither an audio SLB nor a visual reference—rather than from a fundamental limitation of the step count. A short silent audio SLB or a visual guide image are thus alternative mitigations; the I2V experiment provides empirical evidence that the latter is sufficient.

Subsequent chunks in both experiments benefit from the audio SLB and achieve high coherence (≥ 0.81 in S1; ≥ 0.91 by S2 chunks 4–5 in T2V). The I2V Stage 2 audio shows a localised collapse at chunk 3 ($+0.163 \rightarrow +0.175$), tracing back to the anti-correlated audio in Stage 1 chunk 2's second half (-0.824), which Stage 2 refinement only partially corrects.

7.4 Stability Margin

With $\rho_{\text{loop}} = 0.570$, the CLSS feedback loop provides a 43% gain reduction per chunk before accounting for the model's open-loop gain g. The band 3 gain in chunk 3 (0.938) indicates that high-frequency latent energy was growing by $\approx 6.7\%$ per chunk without shrinkage correction—giving a rough empirical estimate of $g \approx 1.07$ for the spectral channel in this run. The resulting $\rho_{\text{closed}} \approx 0.57 \times 1.07 \approx 0.61$ provides ample margin below the stability boundary of 1.

7.5 Multi-Scene Prompting and Identity Metrics

The Stage 2 identity_sim plateau at 0.41–0.52 across chunks 2–5 is consistent with three equally-distinct scenes. Stage 2 lacks an anchor bank; identity tracking is reduced to comparing against the fixed first-chunk reference, which conflates intended scene changes with drift. Adding an anchor bank to Stage 2 would allow scene-aware identity tracking during refinement.

7.6 Measured Failure Modes at Longer Horizons

Extended runs (10 chunks, 43 s) with per-frame telemetry exposed two repetition mechanisms with distinct signatures.

Window-locked audio arcs. The audio energy peak locks to the same relative window position across chunks: 94% of the window at 13 lf chunks (frame 102/109, eight consecutive chunks) and 98% at 31 lf chunks — the arc *scales with the window*, implicating the fed-forward conditioning geometry. Context envelope flattening (Section 5.3) breaks the lock; chunk-to-chunk loudness-gesture correlation nevertheless remains high (0.7–0.95), so repetition is reduced, not eliminated.

A $\sim$4-second intrinsic video oscillation. Coarse scene layout departs from the scene reference and partially returns with a ≈ 4 s wall-clock period that is *invariant to chunk length*: at 13-lf (4.3 s) chunks the dip lands at new-frame 5–7 in 10/10 chunks (appearing seam-locked), while at 31-lf (9.9 s) chunks it appears twice per chunk at the same seconds-period. Context strength (SLB at i2v strength) and context length (overlap 3/5/8) were both tested and neither removes it; the oscillation appears intrinsic to the model's trained temporal horizon. Longer overlap trades the failure shape: at overlap 8 the per-chunk return weakens into monotonic scene departure on some contents.

Stage-2 audio seams. With decoupled Stage-2 audio refinement, a high denoise fraction (σ-scale 1.0) re-rolls $\approx 45\%$ of audio content per window and the re-rolls disagree at every window boundary (boundary cosine 0.32–0.42) *regardless* of the audio SLB length — the SLB pins the seam frame, not the style of the re-roll. At denoise 0.4 the pass polishes instead (seed similarity 0.87–0.92) and seams hold at 0.70–0.83. Rule: high-denoise audio rebuild only as a single window; multi-window refinement at ≤ 0.4.

8 Limitations

Single experimental configuration. All results in this paper are from a single generation run at one set of hyperparameters (new_lf = 31, overlap = 8, 3 chunks, 3 scenes) on one hardware configuration (16 GB VRAM, RTX-class). No baseline comparisons against open-loop streaming are reported, and no ablation of the individual CLSS components (re-noising alone, AdaIN alone, shrinkage alone, anchor bank alone) has been performed. The paper therefore cannot quantify the marginal contribution of each correction or demonstrate improvement over the open-loop baseline from the same metrics. These comparisons are left to future work.

Stability metrics vs. perceptual quality. The reported metrics—boundary cosine similarity, intra-chunk cosine similarity, audio within-chunk segment similarity—measure the *stability* of the latent representation across chunks. They do not assess the perceptual quality of the generated content, prompt adherence, or realism. Standard quality metrics (FID, IS, CLIP similarity, human evaluation) would be required to characterise the generated video as a content generation system. High boundary similarity is a necessary but not sufficient condition for high video quality.

Object-level identity is unobserved. Every reported metric is computed from mean-pooled channel features or low-band spatial maps and is therefore *object-blind*: a vehicle or person can be replaced by a different instance across a chunk boundary while boundary and identity cosines remain above 0.95. This failure class was observed perceptually and is invisible to the telemetry. High-similarity readings must not be interpreted as object persistence. The available mitigation is textual: one prompt per chunk via `CLSSScenePrompts`, re-describing the persistent objects explicitly. Object-level tracking metrics (detection/re-identification on decoded frames) are future work.

Stability model assumptions. The linear gain model in Section 3.5 assumes that g is approximately constant and that the AdaIN and re-noising corrections act as independent multiplicative attenuators. Both assumptions simplify the analysis: the actual closed-loop dynamics involve higher-order interactions between corrections, guidance, and scene content. A more rigorous treatment (e.g., Lyapunov stability for the latent iteration) would require characterising the denoiser's Jacobian across a distribution of latent inputs.

Hardware specificity. The GGUF 4-bit quantisation and CPU block-streaming configuration was designed for a specific VRAM budget (16 GB). Performance, latency, and numerical behaviour may differ on GPUs with more VRAM (where block-streaming is unnecessary) or with different quantisation schemes (e.g., Q8 or FP8).

9 Related Work

Long video generation. StreamingT2V [1] and FIFO-Diffusion [2] generate long videos by sliding-window denoising. Gen-L-Video [3] uses temporal attention patching. CLSS differs by applying closed-loop corrections specifically designed for the latent-streaming regime of large audio-video models.

Exposure bias in diffusion. The exposure bias problem in autoregressive models is well-studied [4]. For diffusion models, Ning et al. [5] and Salimans & Ho [6] address related distributional mismatch issues during training, but inference-time corrections for streaming have received less attention.

Statistical normalisation for consistency. Adaptive Instance Normalisation (AdaIN) [7] was introduced for style transfer, where the goal is to match the mean and standard deviation of a content feature map to those of a style reference. EMA-based statistical normalisation across video frames has also appeared in temporal consistency methods. In CLSS, AdaIN is repurposed as a *drift correction* applied to denoised latents between chunks, using the cross-chunk EMA as the style reference. The combination of EMA tracking with AdaIN blending (rather than full replacement) is motivated by the streaming setting, where the reference must itself evolve to allow intentional scene changes.

Frequency-domain video editing. Yang et al. [8] and Ceylan et al. [9] use frequency-domain constraints for temporal consistency in video editing. CLSS applies frequency shrinkage in the *latent* domain to a streaming generation setting, targeting energy accumulation rather than pixel-domain flicker.

10 Conclusion

We presented CLSS, a four-component closed-loop streaming synthesis framework that enables stable long audio-video generation with LTX-2.3 22B on consumer 16 GB VRAM hardware. The four corrections—calibrated context re-noising, EMA-tracked AdaIN, frequency-band soft shrinkage, and a dynamic anchor bank— achieve $\rho_{\text{loop}} = 0.570$ and maintain Stage 1 boundary similarities above 0.946 across both T2V and I2V experiments. A two-stage pipeline (streaming S1 + chunked distilled S2) delivers a fully refined audio-video output in approximately 55–56 minutes per 30-second clip on a single 16 GB GPU.

Comparison of the T2V and I2V experiments reveals that audio coherence in the first generated chunk—where no audio SLB is available—depends on the presence of a strong visual conditioning anchor, not solely on the audio SLB. T2V chunk 1 audio is anti-correlated (≈ -0.3 per segment pair) while I2V chunk 1 audio, conditioned on a guide image, achieves $\approx +0.999$. This indicates that the joint audio-video cross-modal attention in LTX-2.3 provides sufficient visual grounding to maintain audio coherence when a visual reference is present. A natural extension is to supply a short audio reference or visual guide before chunk 1 even in T2V mode, priming the pipeline's audio coherence from the start.

The July 2026 audit is itself a result. Two defects — unseeded Stage-1 audio noise and a guidance stack silently disabled by ComfyUI's packed-latent sampling representation — survived weeks of experimentation because both failed *silently* and the metrics they corrupted still moved plausibly. The telemetry that finally exposed them (unconditional settings dumps, one-time guidance-path announcements, and treating wall-clock invariance as evidence of a missing pass) is now part of the pipeline, and the reference-parity guider (split CFG, modality guidance, STG) is verified active; the first fully guided run is the most stationary measured to date.

The ComfyUI node integration with per-chunk diagnostic telemetry supports rapid empirical validation of CLSS hyperparameter choices and provides interpretable monitoring of the feedback loop during generation.

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References

[1] R. Henschel, L. Khachatryan, D. Hayrapetyan, H. Poghosyan, R. Abrahamyan, A. Nersisyan, S. I. Yeranosyan, and O. Kacher. StreamingT2V: Consistent, dynamic, and extendable long video generation from text. *arXiv:2403.14773*, 2024.

[2] J. Kim, J. Jeong, S. Park, M. Kim, S. Kim, and J. Ye. FIFO-Diffusion: Generating infinite videos from text without training. *arXiv:2405.11473*, 2024.

[3] F. Wang, W. Chen, G. Song, H. Zhang, W. Li, C. Shen, and T. Mei. Gen-L-Video: Multi-text conditioned long video generation via temporal co-denoising. *arXiv:2305.18264*, 2023.

[4] S. Bengio, O. Vinyals, N. Jaitly, and N. Shazeer. Scheduled sampling for sequence prediction with recurrent neural networks. In *NeurIPS*, 2015.

[5] M. Ning, M. Li, J. Su, A. A. Salah, and I. Ertugrul. Elucidating the exposure bias in diffusion models. *arXiv:2308.15321*, 2023.

[6] T. Salimans and J. Ho. Progressive distillation for fast sampling of diffusion models. In *ICLR*, 2022.

[7] X. Huang and S. Belongie. Arbitrary style transfer in real-time with adaptive instance normalization. In *ICCV*, 2017.

[8] S. Yang, Y. Zhou, Z. Liu, and C. C. Loy. FRESCO: Spatial-temporal correspondence for zero-shot video translation. In *CVPR*, 2024.

[9] D. Ceylan, C.-H. Huang, and N. J. Mitra. Pix2Video: Video editing using image diffusion. In *ICCV*, 2023.